DarkIR-GAN: GAN-Based Low-Light Blurred Image Restoration

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***Abstract*—Low illumination and motion blur severely constrain visual perception in extreme environments. Existing reconstruction-oriented networks often suppress fine local structure when minimizing distortion losses. This paper proposes DarkIR-GAN, a generative adversarial framework for joint low-light enhancement and deblurring. It retains the DarkIR generator and its reconstruction objective, and adds only a PatchGAN discriminator with least-squares adversarial supervision. The baseline-continued and adversarial branches restore the same Stage-II generator, optimizer, and scheduler states and are each trained for 30 additional epochs. On the complete 1,800-pair LOL-Blur test set, the two 30-epoch endpoints yield LPIPS values of 0.1530 and 0.1301, MUSIQ scores of 55.16 and 60.72, NRQM scores of 5.10 and 5.98, and NIQE values of 5.366 and 4.737 for the baseline-continued and adversarial branches, respectively. PSNR changes from 24.80 to 24.94 dB, while SSIM changes from 0.8558 to 0.8553. The results indicate improved perceptual-quality metrics with nearly unchanged distortion fidelity in the reported experiment.**

***Index Terms*—low-light blurred images, image restoration, generative adversarial networks, perceptual quality assessment**

I. Introduction and Related Work

Imaging under extreme conditions—dimly lit industrial facilities, underground tunnels, and nighttime surveillance—frequently suffers from compound degradation where low illumination and motion blur coexist. Long exposure under low light introduces motion blur from camera shake or scene dynamics and can also degrade downstream perception tasks such as object detection [1]. The multiplicative coupling of blur and illumination renders joint restoration fundamentally ill-posed.

Low-light enhancement has progressed from Retinex-based methods [2], [3] to deep learning approaches including early Convolutional Neural Network (CNN)-based methods [4], Zero-DCE [5], KinD [6], SCI [7], and Transformer-based Retinexformer [8]. Image deblurring has similarly advanced from hand-crafted priors [9] to multi-scale CNNs including MIMO [10], [11], Transformer-based Restormer [12], and modern CNN baselines such as NAFNet [13]. Other relevant restoration baselines include LLFlow [14], while dedicated joint low-light enhancement and deblurring methods include LEDNet [15] and LIEDNet [16]. Dong et al. [17] recently introduced a zero-reference visual-autoregressive approach to the same joint restoration task without paired supervision.

However, these distortion-minimizing methods optimize L_1/L_2 losses that produce over-smoothed outputs—a consequence of the perception-distortion tradeoff [18]. Generative Adversarial Network (GAN)-based methods mitigate this by providing implicit high-frequency supervision, achieving notable perceptual gains in super-resolution (SRGAN [19], ESRGAN [20]) and deblurring (DeblurGAN [1]). Building upon this, PatchGAN discriminators [21] offer localized adversarial feedback that is particularly effective for texture-level discrimination in restoration tasks. Yet the application of adversarial learning to the joint low-light enhancement and deblurring task remains largely unexplored. The DarkIR architecture [22] provides a robust backbone with multi-scale skip connections, but its non-adversarial composite reconstruction and perceptual objective does not explicitly enforce distribution-level realism. To bridge this gap, we propose DarkIR-GAN, which introduces adversarial supervision into the DarkIR backbone. Our contributions are:

- Integrating a PatchGAN discriminator into the DarkIR generator to form DarkIR-GAN, encouraging locally realistic high-frequency statistics via adversarial supervision.
- A three-stage protocol (domain adaptation, baseline convergence, and matched continuation) in which the two final branches share the generator architecture, initialization, optimizer state, scheduler state, reconstruction objective, data configuration, and training budget.
- A completed 30-epoch comparison on LOL-Blur [15] showing improvements in full-reference LPIPS and three no-reference measures (MUSIQ, NIQE, and NRQM), while final-checkpoint Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) remain numerically close to the continued baseline.

II. DarkIR-GAN Restoration Method

A. Problem Formulation and Preliminaries

Given $I_{\text{degraded}} \in \mathbb{R}^{H\times W\times 3}$ under $I_{\text{degraded}} = (I_{\text{sharp}} * k) \cdot L + n$, the objective is to recover I_{sharp}. The coupled effects of blur k, illumination L, and additive noise n render this problem severely ill-posed. Pixel-wise reconstruction objectives tend to favor conservative solutions under ambiguous restoration targets, often suppressing fine high-frequency details [18]. Adversarial learning [23] provides complementary distributional supervision that encourages locally realistic image statistics. We adopt Least-Squares GAN (LSGAN) [24] for its smoother gradient landscape and training stability. The

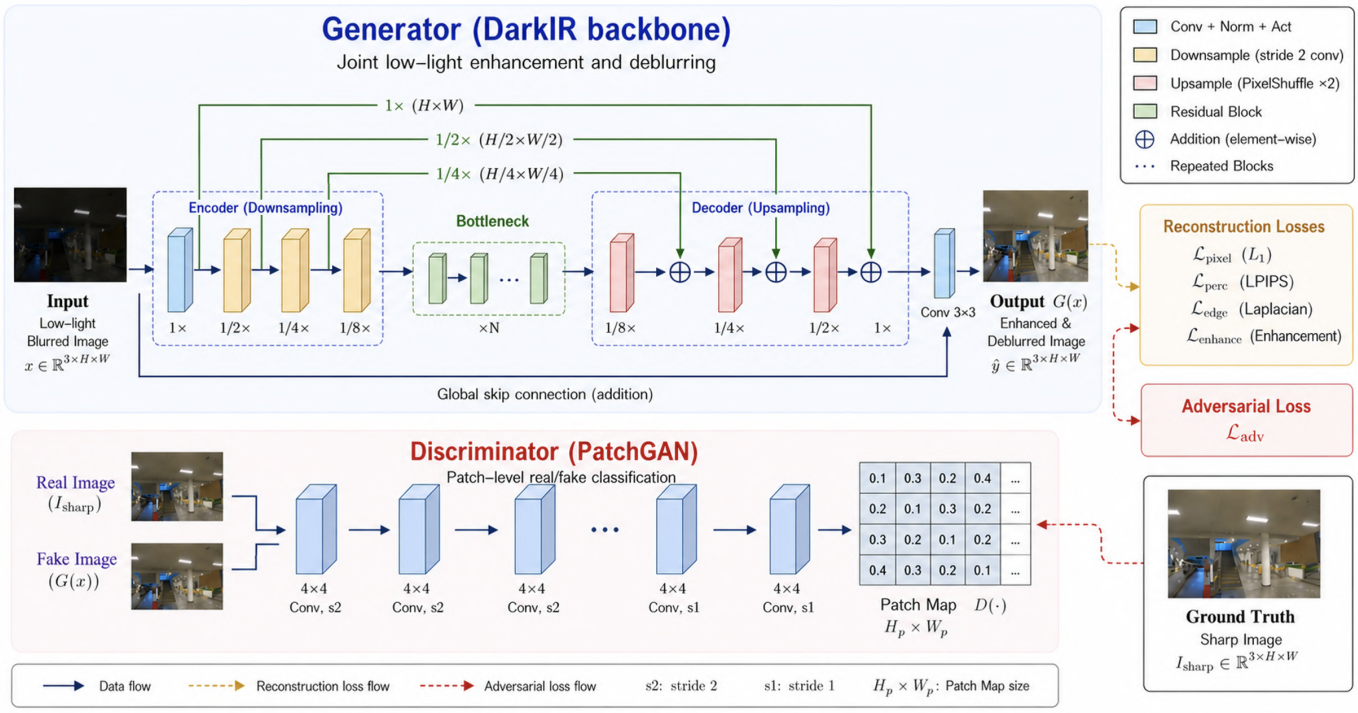


Fig. 1. DarkIR-GAN overall architecture. The generator performs low-light enhancement and deblurring, while the PatchGAN discriminator provides adversarial supervision.

proposed framework jointly optimizes G and D with a hybrid loss (Section II-C), where G performs joint enhancement and deblurring and D provides local adversarial supervision on texture realism.

B. Overall Architecture

The DarkIR-GAN framework, illustrated in Figure 1, adopts a U-Net-style generator-discriminator architecture tailored for compound low-light and blur degradation. The generator G follows an encoder-decoder topology with multi-scale skip connections, enabling both global context aggregation and fine-grained structural preservation. The discriminator D employs a fully convolutional PatchGAN design [21] that operates on local image patches to provide spatially dense adversarial feedback.

1) Generator Design: The generator is the unchanged DarkIR implementation [22]. A 3×3 introductory convolution projects an RGB image to $C{=}64$ channels. Three encoder stages contain $[2, 2, 4]$ blocks and double the width ($64\to128\to256\to512$) through stride-2 convolutions. One encoder and one decoder bottleneck block operate at the lowest resolution. Three PixelShuffle stages with $[2, 2, 2]$ decoder blocks restore the original resolution, using decoder dilations $[1, 2, 3]$. Additive encoder–decoder skips and a global input residual preserve spatial structure. The implementation pads dimensions to multiples of eight and crops the prediction back to the native image size; the formal runs therefore use the native 1120×640 LOL-Blur frames rather than a 256×256 resize.

2) Discriminator Design: The PatchGAN discriminator [21] provides localized high-frequency supervision. It uses five 4×4 convolutions with channels $3\to64\to128\to256\to512\to1$ and strides $[2, 2, 2, 1, 1]$. The first four convolutions are followed by Leaky Rectified Linear Unit (LeakyReLU) activations with slope 0.2; batch normalization is applied after the second, third, and fourth convolutions. The resulting patch map supplies spatially dense real/fake feedback.

C. Hybrid Objective Function

Before loss evaluation, the generator output and target are clamped to $[0, 1]$. The implemented generator objective preserves the continued-baseline reconstruction terms exactly and adds only the adversarial term:

$$\begin{aligned}\mathcal{L}_G =& \mathcal{L}_{\text{pixel}} + 0.01\mathcal{L}_{\text{LPIPS}} + 25\mathcal{L}_{\text{edge}} \\ &+ 0.25\mathcal{L}_{\text{enhance}} + 0.01\mathcal{L}_{\text{adv}}.\end{aligned} \tag{1}$$

The effective coefficients above include both the loss-object weights in the configuration (0.1, 5.0, and 0.5) and the outer multipliers used by the baseline training loop (0.1, 5.0, and 0.5). This two-level weighting is retained deliberately so that adversarial supervision is the only generator-side change.

Pixel Loss. L_1 reconstruction anchors the output to the ground truth:

$$\mathcal{L}_{\text{pixel}} = \|G(I_{\text{degraded}}) - I_{\text{sharp}}\|_1. \tag{2}$$

Perceptual Loss. The baseline uses the VGG-based LPIPS implementation (LPIPS-VGG) introduced in [25]:

$$\mathcal{L}_{\text{LPIPS}} = \text{LPIPS}_{\text{VGG}}\big(G(I_{\text{degraded}}), I_{\text{sharp}}\big). \tag{3}$$

Edge Loss. An edge-preserving loss based on a Laplacian-pyramid residual improves boundary sharpness:

$$\mathcal{L}_{\text{edge}} = \mathcal{L}_{\text{lap}}\big(G(I_{\text{degraded}}),\, I_{\text{sharp}}\big). \tag{4}$$

Enhance Loss. The continued baseline applies its enhancement loss to the generator output and target. It combines image-domain L_1 distance with VGG19 feature-domain L_1 distance:

$$\begin{aligned}\mathcal{L}_{\text{enhance}} = \|G(I_{\text{degraded}}) - I_{\text{sharp}}\|_1 \\ + 0.01 \sum_l \|\phi_l(G(I_{\text{degraded}})) - \phi_l(I_{\text{sharp}})\|_1.\end{aligned} \tag{5}$$

Adversarial Loss. Based on LSGAN [24] with label smoothing (real target $= 0.9$), the discriminator and generator adversarial losses are:

$$\mathcal{L}_D = \frac{1}{2}\big(\|D(I_{\text{sharp}}) - 0.9\|_2^2 + \|D(G(I_{\text{degraded}}))\|_2^2\big), \tag{6}$$

$$\mathcal{L}_{\text{adv}} = \|D(G(I_{\text{degraded}})) - 1\|_2^2. \tag{7}$$

While the reconstruction losses collectively anchor the output in the correct structural and chromatic space, $\mathcal{L}_{\text{adv}}$ adds a gradient signal that encourages local high-frequency statistics toward the target-image distribution.

D. Three-Stage Training Protocol

Stage III restores both branches from the same Stage-II checkpoint, including the generator, AdamW optimizer, and cosine-scheduler states. The restored generator learning rate is 10^{-6} at the Stage-III start. Baseline-continued retains reconstruction training only, whereas DarkIR-GAN adds the randomly initialized discriminator and adversarial term. Both branches use the same dataset split, data-loading configuration, and number of generator epochs and updates.

Algorithm 1 Three-Stage Training Protocol for DarkIR-GAN

Require: Full LOL-Blur training set $\mathcal{D}_{\text{full}}$ (10,200 paired samples);
Require: Sampled subset $\mathcal{D}_{\text{sub}}$ (2,550 paired samples, 15 per scene);
Require: Pre-trained DarkIR weights θ_G^0; learning rate schedules η_G, η_D.
// Stage I: Full-Data Domain Adaptation
Load generator weights $\theta_G \leftarrow \theta_G^0$
Train G on $\mathcal{D}_{\text{full}}$ for 1 epoch with reconstruction losses only
Obtain domain-adapted checkpoint θ_G^{I}
// Stage II: Baseline Convergence Training
Initialize G from θ_G^{I}, set $\eta_G = 1 \times 10^{-4}$
for epoch $= 1$ **to** 30 **do**
  Train G on $\mathcal{D}_{\text{sub}}$ using $\mathcal{L}_{\text{pixel}} + \mathcal{L}_{\text{perc}} + \mathcal{L}_{\text{edge}} + \mathcal{L}_{\text{enhance}}$
  Evaluate on the test set every 3 epochs
end for
Save converged checkpoint θ_G^{II}
// Stage III: Synchronized Comparative Fine-Tuning
Fork 1 – Baseline-continued:
  Restore G, AdamW, and scheduler from Stage II (initial $\eta_G = 10^{-6}$)
  for epoch $= 1$ **to** 30 **do**
    Train G on $\mathcal{D}_{\text{sub}}$ with reconstruction losses only using the restored scheduler
  end for
Fork 2 – DarkIR-GAN:
  Restore the same G/optimizer/scheduler states; randomly initialize D with $\eta_D = 10^{-4}$
  for epoch $= 1$ **to** 30 **do**
    Update D with $\mathcal{L}_D$, then G with $\mathcal{L}_G$; step both schedulers
  end for
Ensure: Restored outputs $G(I_{\text{degraded}})$ from both forks for comparison.

III. Experiments and Results

A. Experimental Setup

The completed runs use an NVIDIA GeForce RTX 4080 SUPER, Python 3.10.12, PyTorch 2.5.1 with CUDA 12.4, native 1120×640 images, and batch size 1. Training samples 15 paired frames from each scene (2,550 paired samples per epoch); test-set evaluation uses all 1,800 paired frames every three epochs. The generator uses AdamW [26] with $(\beta_1, \beta_2) = (0.9, 0.999)$ and weight decay 10^{-5}; both branches restore the same optimizer and cosine-scheduler states, starting Stage III at 10^{-6} and stepping the restored scheduler throughout the 30-epoch continuation. The discriminator uses Adam with $(0.5, 0.999)$, learning rate 10^{-4}, and cosine annealing [27] to 10^{-6}. We report PSNR, SSIM [28], LPIPS [25], MUSIQ [29], NIQE [30], and NRQM [31]. Originally developed for super-resolution, NRQM is used only as an auxiliary cross-domain no-reference indicator here.

B. Stage-Wise Results

The three-stage progression is summarized in Table I. The completed Stage-III rows use the final epoch-30 checkpoints from both 30-epoch continuations, avoiding mixed checkpoint-selection criteria. Relative to Baseline-continued, DarkIR-GAN improves PSNR by 0.137 dB and changes SSIM by only -0.0005. The perceptual gains are larger: LPIPS decreases by 0.0229 (14.97%), MUSIQ increases by 5.555 (10.07%), NIQE decreases by 0.629 (11.72%), and NRQM increases by 0.881 (17.27%). Moreover, all ten scheduled test-set evaluations favor DarkIR-GAN on LPIPS, MUSIQ, NIQE, and NRQM. Baseline-continued retains the higher peak PSNR across training (25.468 vs. 25.284 dB), indicating that the adversarial objective primarily changes the perceptual operating point rather than uniformly improving every fidelity measure.

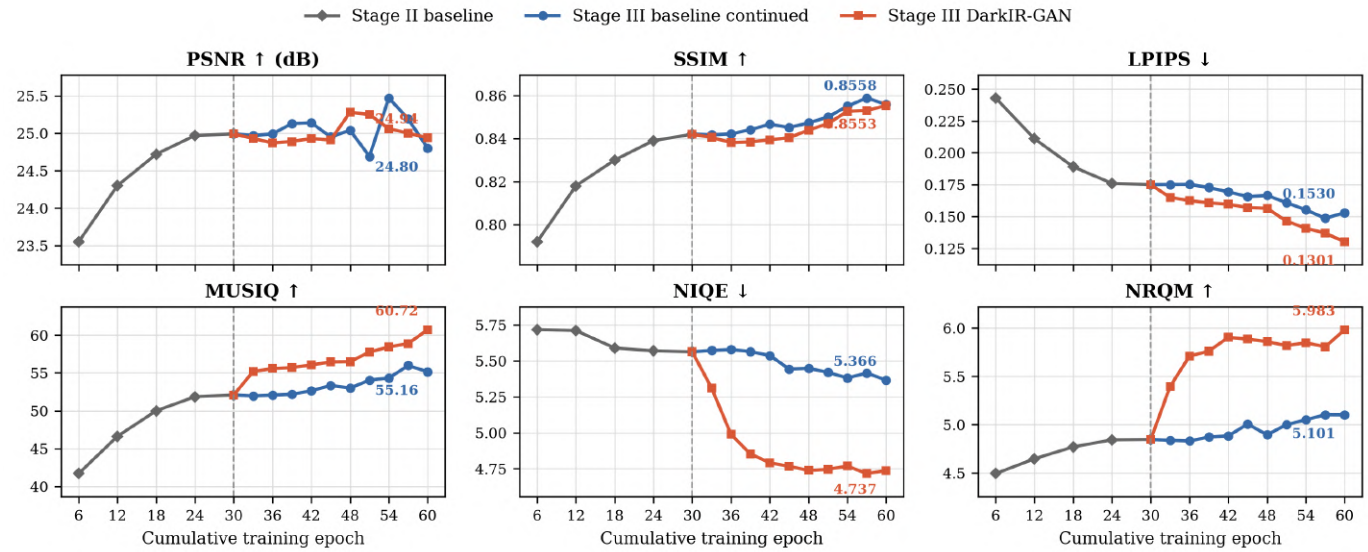


Fig. 2. Stage-II baseline convergence and completed Stage-III test-set evaluation trajectories on all 1,800 test pairs. Stage II is shown at six-epoch intervals; Stage III is evaluated every three epochs after the shared epoch-30 fork. The horizontal axis reports cumulative training epochs, the dashed line marks the stage boundary, and arrows indicate the preferred direction.



Fig. 3. Representative qualitative behavior (from left to right: input, continued baseline, and DarkIR-GAN). Final matched quantitative results are reported in Table I.

TABLE I
STAGE-WISE TRAINING PROGRESSION AND FINAL COMPARISON. PSNR/SSIM: DISTORTION; LPIPS: FULL-REFERENCE PERCEPTUAL; MUSIQ/NIQE/NRQM: NO-REFERENCE PERCEPTUAL.

	PSNR↑	SSIM↑	MUSIQ↑	LPIPS↓	NIQE↓	NRQM↑
Stage I: Domain Adaptation (1 epoch, 10,200 paired samples)						
After adaptation	22.56	0.762	37.41	0.256	5.730	4.571
Stage II: Baseline Convergence (30 epochs, 2,550 paired samples)						
6 epochs	23.55	0.792	41.70	0.243	5.716	4.494
12 epochs	24.30	0.818	46.62	0.211	5.710	4.647
18 epochs	24.72	0.830	50.00	0.189	5.589	4.770
24 epochs	24.97	0.839	51.85	0.176	5.569	4.843
30 epochs	24.99	0.842	52.09	0.175	5.562	4.847
Stage III: Comparative Fine-Tuning (30 epochs)						
Baseline-continued	24.803	**0.8558**	55.163	0.1530	5.366	5.102
DarkIR-GAN (Ours)	**24.940**	0.8553	**60.718**	**0.1301**	**4.737**	**5.983**
Δ (GAN − Base)	+0.137	−0.0005	+5.555	−0.0229	−0.629	+0.881

TABLE II
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS FOR JOINT LOW-LIGHT ENHANCEMENT AND DEBLURRING. BEST IN **BOLD**, SECOND-BEST UNDERLINED.

Method (Venue, Year)	PSNR↑	SSIM↑	MUSIQ↑	LPIPS↓	NIQE↓	NRQM↑
Cascaded Two-stage Pipeline						
Restormer → LLFlow	21.89	0.772	35.42	0.347	4.982	5.011
General Restoration / Enhancement Baselines						
MIMO (ICCV '21) [11]	22.41	0.835	35.37	0.262	4.851	5.140
Restormer (CVPR '22) [12]	23.63	0.841	36.65	0.247	5.093	5.497
LLFlow (AAAI '22) [14]	24.48	0.846	34.87	0.235	5.202	5.312
Joint End-to-end Low-light Deblurring						
LEDNet (ECCV '22) [15]	26.06	0.852	43.48	0.217	5.435	5.421
LIEDNet (TCSVT '25) [16]	27.47	0.884	44.75	0.183	5.129	5.532
LIEDNet-L (TCSVT '25) [16]	**27.99**	**0.897**	44.49	0.161	4.806	5.808
DarkIR (Baseline-continued)	24.803	0.8558	55.163	0.1530	5.366	5.102
Proposed Perception-oriented Variant						
DarkIR-GAN (Ours)	24.940	0.8553	**60.718**	**0.1301**	**4.737**	**5.983**

C. Comparison with State-of-the-Art Methods

To contextualize DarkIR-GAN, we compare it with reported results on LOL-Blur. The methods are grouped into a cascaded pipeline, general restoration/enhancement baselines, dedicated joint low-light deblurring methods, and the proposed perception-oriented variant. For competing methods, we report previously published LOL-Blur benchmark results without re-evaluation. Our two rows use the completed native-resolution evaluations described above.

As shown in Table II, LIEDNet-L retains the highest reported PSNR (27.99 dB) and SSIM (0.897). Baseline-continued reaches 0.8558 SSIM, above LEDNet but below the LIEDNet family. The cascaded Restormer→LLFlow pipeline remains substantially weaker on distortion and LPIPS, sup-

porting end-to-end joint restoration.

In the perception-oriented category, DarkIR-GAN obtains the strongest listed LPIPS, MUSIQ, NIQE, and NRQM values. Relative to LIEDNet-L, it increases MUSIQ by 16.23, reduces LPIPS by 0.031, reduces NIQE by 0.069, and increases NRQM by 0.175. The LIEDNet variants retain the strongest PSNR/SSIM, consistent with different operating points on the perception-distortion curve [18].

D. Discussion

Overall, DarkIR-GAN shifts the reported metrics toward perceptual quality without a material final-checkpoint loss in distortion fidelity. The saved examples show stronger local contrast and more pronounced edges, while fine text and high-contrast boundaries remain challenging and may exhibit false contours or color ringing. The experiment compares two continuation configurations initialized from the same pre-trained state and evaluated under the same protocol. Additional repeated runs and paired per-image analyses would further characterize the variability of the observed differences.

IV. Conclusion and Future Work

This paper presented DarkIR-GAN as an adversarial continuation of DarkIR for low-light blurred image restoration. At the two 30-epoch endpoints, the DarkIR-GAN branch obtains better LPIPS and all three no-reference quality metrics, together with slightly higher PSNR and nearly unchanged SSIM. The same metric direction appears across the ten scheduled evaluations, indicating that the endpoint observation is also reflected along the reported training trajectory.

The discriminator increases training time (32.84 vs. 30.80 hours) but not inference cost because it is discarded at deployment. Current limitations include edge artifacts, experimental variability, and the synthetic-camera domain gap. Future work will include repeated runs, paired statistics, and real nighttime mobile imagery.

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